



Fig. 5: Jeedom



Fig. 6: CalaOS

automatically connects via a QR code without the need for any configuration. You will find on your mobile the same functionality to your Jeedom (scenarios, connected devices and home automation, plugins) and you can also customise your application with shortcuts and more.

The main features of Jeedom are:

- Manage the safety of goods and people.
- Automate heating for better comfort and energy savings.
- View and manage energy use to anticipate expenses and reduce usage.
- Communicate by voice, SMS, e-mail or mobile applications.
- Manage all of the home's automatic devices: shutters, gate, lights.
- Manage multimedia audio and video devices and connected objects.

Jeedom includes its own Market, like what you find on your smartphones with the App Store or Play Store. It allows you to add home automation features, assures compatibility

with new home automation modules and even lets you customise its fixed and mobile interfaces.

3. CalaOS. This FOSS lets you control and monitor your home with many features. It can be easily installed and used to transform your home into a smart home. CalaOS has a long background in home automation and is built in several different layers. It has low-level as well as uniform and powerful user interfaces. It knows how to talk to hardware and gives you the convenience to automate your home easily.

It was first developed by a French company of the same name but when the company was closed in 2013, the entire code was open sourced and released as GNU General Public License. A small community started around the project to continue the development and is growing every day.

CalaOS can be installed on Linux, Mac, Windows, iOS, and Android operating systems. The iOS and Android apps can be used for home automation using a smartphone, remotely.

Initially, CalaOS supported a special kind of hardware called programmable logic controller (PLC) to automate a building or apartment. PLCs can work as a controller to control various home devices like wall switches, fans, air-conditioners, water heaters, etc, but have to be wired to the PLC. However, the newer version of CalaOS also supports GPIO, Zibase, and IoT devices.

The main features of CalaOS are:

Hardware support. CalaOS supports hardware like Wago PLC, Raspberry Pi, Zodianet's ZiBASE, Cubieboard, Squeezebox, and CCTV. The develop-

ment community is improving the app to support more hardware arriving in the market. CalaOS is very easy to install; just by burning a memory card, it is possible to have a complete working software installation.

Software stack. CalaOS comes with the following software stack:

- CalaOS Server: Core application.
- CalaOS Home: Touchscreen user interface.
- CalaOS Webapp: HTML5 web application.
- CalaOS Installer: Configuration tool.
- CalaOS iOS: iOS App.
- CalaOS Android: Android App.
- CalaOS-OS: Full Linux OS with everything preconfigured.
- Do It Yourself (DIY): With CalaOS, one can quickly and easily install a complete installation. One can get support from the online community regarding any technical issues. The Wiki is full of documentation on how it works and the CalaOS forum can help answer any specific question. One can configure the app as required and, as it is an OSS, it is even possible to modify.

Apart from the above three software, there are many more OSS available for home automation but most of them do not support iOS and Android OS for using on a smartphone. Few more OSS that can be used for home automation are Home Assistant, Domoticz, OpenMotics, LinuxMCE, PiDome, MisterHouse, and Smarthomatic.

From 'home sweet home' to 'home smart home'

With the help of proper hardware and software, it is possible to make the home automation more secure and smart. Home automation also saves overall power consumption and helps reduce the electricity bill. As technology improves and more advanced hardware/software comes to the market—home automation will become easier and more convenient. Someday, a home sweet home will also become a home smart home for everyone. **EFY**

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Debojit Acharjee is software engineer/writer